An Arduino-Based Waterlogging Detection and Barrier-Control Prototype for Subterranean Infrastructure

System Architecture and Validation Methodology

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ABSTRACT

Waterlogging in road underpasses and other low-lying infrastructure can create serious mobility and safety risks when rising water is not detected promptly. This paper documents an Arduino-based, bench-scale prototype intended to estimate water depth, display local status, and demonstrate a servo-operated model barrier. The reported components are an Arduino Uno/ATmega328P, an HC-SR04 ultrasonic sensor, and a 20 × 4 LCD with an I²C interface. Depth is estimated from sensor-to-surface distance and a calibrated reference height. A proposed validation baseline specifies invalid-echo handling, threshold classification, event logging, and conditional actuator-feedback checks. The supplied photographs demonstrate a physical scaled model, but the available materials do not include firmware, an as-built circuit diagram, calibration records, raw readings, repeated-trial results, or measured detection and actuation latency. Consequently, sensor accuracy, pump efficiency, remote alerts, automatic dewatering, and independent hardware-override performance cannot be treated as verified outcomes. The contribution is a reproducible evaluation framework and a clear boundary between reported components and proposed extensions. Controlled testing and independent safety assessment are required before operational deployment.

Keywords: *waterlogging detection; ultrasonic sensing; Arduino Uno; embedded control; barrier actuation; subterranean infrastructure*

I. INTRODUCTION

Waterlogging at road underpasses, subway entrances, service passages, and other low-lying locations can rapidly turn a transport route into a hazardous area. Water may rise faster than staff can inspect a site, particularly during intense rainfall or drainage-system failure. A useful monitoring arrangement should acquire a water-level indicator, interpret it against site-specific thresholds, communicate the local state clearly, and support timely access-control or evacuation decisions. Remote telemetry may help maintenance teams coordinate a response where connectivity is available. These functions are related but distinct: local sensing and actuation do not, by themselves, establish Internet connectivity or a validated flood-warning service.

The Internet of Things (IoT) integrates sensing, embedded processing, communications, and service layers; its value depends on reliable integration rather than the presence of a microcontroller alone [1]–[3]. Water-level and flood-warning studies have explored ultrasonic ranging, cellular messages, low-power wide-area networking, cloud dashboards, and multi-sensor monitoring [4]–[13]. Tunnel-focused research also emphasizes flood propagation and timely, scenario-aware warning [14]–[16]. For a small educational prototype, the first engineering task is to document what is actually implemented and then determine experimentally whether its behaviour is adequate for the intended purpose.

The supplied project description identifies an Arduino Uno/ATmega328P, an HC-SR04 ultrasonic sensor, a servo-operated model barrier, and a 20 × 4 I²C LCD. Four photographs show a physical mock-up during a demonstration. However, the available materials do not include source code, a complete circuit diagram, calibration sheets, numerical test logs, or identifiable GSM, Wi-Fi, or LoRaWAN hardware. A pump, high-current relay, and hardware override are discussed, but their implementation and testing are not documented sufficiently to treat them as demonstrated subsystems.

This paper contributes an evidence-bounded description of the reported local prototype; a focused synthesis of water-level sensing, flood-warning communication, and underground-flood research; a distinction between documented hardware and proposed extensions; and a measurement protocol for future validation. It is a prototype report and evaluation framework, not a field-performance study.

II. RELATED WORK AND LITERATURE REVIEW

A. IoT sensing and actuation architectures

Atzori, Iera, and Morabito describe IoT as an integration of identification, sensor/actuator networks, communications, and distributed intelligence [1]. Gubbi et al. discuss the convergence of wireless sensing, cloud resources, and connected actuation [2], while Zanella et al. examine urban IoT architectures and their role in smart-city services [3]. These foundations matter because a flood-management system requires a documented path from sensor acquisition to local decision-making and, when described as IoT-enabled, a verified communications path to external services. An LCD supports local visibility but does not

prove remote telemetry.

B. Water-level sensing and flood-warning prototypes

Low-cost water-level systems commonly connect ultrasonic sensors to embedded controllers. Menaka and Shrinithi describe an IoT-linked monitor with a network-capable controller and application interface [4]. Subekti et al. examine the static characteristics of the HC-SR04 for Arduino-based water-level monitoring, highlighting the need to measure sensor behaviour rather than assume a nominal accuracy [5]. Zakaria, Jabbar, and Sulaiman report an HC-SR04/Arduino sensing unit with LoRaWAN telemetry, dashboards, and field monitoring [6]. Cherif et al. present LevelWAN, an open-source water-level monitoring platform designed for dynamic aquatic environments [7]. These studies provide design references, but their reported performance cannot be transferred to this prototype because geometry, mounting, filtering, sensors, and communications differ.

Cellular communication has also been used for flood warnings. Danang et al. describe an Arduino-based early-warning and monitoring information system [8]. Zain et al. report GSM-based SMS/call alerts using Arduino and ultrasonic sensing [9]. Vijayalakshmi et al. discuss water-level monitoring with Arduino, GSM, and cloud-based presentation [10], while Kaphungkui et al. describe GSM-based monitoring with defined risk levels [11]. Remote notification requires communication hardware, a working network, configured recipients or endpoints, and message-delivery tests; it cannot be inferred from an LCD or an Arduino board alone.

The reviews by Praveen and Vadivel [12] and Pahuriray and Cerna [13] synthesize IoT-based flood-monitoring architectures and warning methods. Because the reviewed systems use different hardware, field conditions, and validation protocols, the literature supports comparative evaluation rather than providing performance evidence for the present model.

C. Underground flooding and tunnel monitoring

Tunnel water ingress presents a different operational context from a household tank or river gauge. Wang et al. propose real-time monitoring and information-fusion warning for water inrush during tunnel construction [14]. Dong et al. use scale-model experiments to study flood patterns and early warning in subway tunnels [15]. Zhang et al. describe multi-source imagery and three-dimensional digital-twin reconstruction for tunnel-waterlogging monitoring [16]. These approaches range from embedded level sensors to information fusion and vision-based reconstruction. Their cost, calibration needs, and computational complexity differ; numerical comparison with a small bench-scale model would be inappropriate without common metrics and equivalent test conditions.

D. Synthesis and research gap

The literature already covers ultrasonic water-level sensing, cellular warning, LoRaWAN monitoring, and experimental tunnel-flood analysis [5]–[16]. It would therefore be inaccurate to claim that automated water-level sensing or tunnel flood warning is itself novel. The narrower contribution here is implementation-oriented: documenting a low-cost scaled model that is intended to combine local distance measurement, a visible status display, and a model barrier, and defining a repeatable validation procedure. Whether the threshold logic and barrier response provide dependable lead time remains an open experimental question.

Table 1. Selected literature and its relevance to the reported prototype.

Study	Approach reported in the source	Relevance and comparison boundary
Subekti et al. [5] (2022)	HC-SR04 characterization and Arduino-based water-level control.	Supports the need for calibration; published accuracy values are not transferable to this model.
Zakaria et al. [6] (2023)	HC-SR04 + Arduino, LoRaWAN gateway, online dashboards, and field monitoring.	Shows what a networked warning path requires; such network hardware is not verified here.
Cherif et al. [7] (2025)	Open-source IoT water-level monitoring for dynamic aquatic environments.	Supports reproducible instrumentation and data collection as evaluation priorities.
Zain et al. [9] (2020)	Arduino, HC-SR04, and GSM-based SMS/call flood warning.	Remote alerting is a distinct module that requires delivery-latency testing.
Wang et al. [14] (2021)	Fusion-based real-time monitoring and early warning for tunnel water inrush.	Highlights the need for robust monitoring beyond a single threshold sensor.
Dong et al. [15] (2024)	Scale-model experiments on flood patterns and warning in subway tunnels.	Relevant infrastructure context; detailed calibration remains necessary here.
Zhang et al. [16] (2026)	Multi-source sensing and digital-twin reconstruction for subway-tunnel waterlogging.	Illustrates an advanced alternative, not a like-for-like low-cost comparator.

III. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

The scaled model is intended to demonstrate whether changes in water-surface height can be sensed and used to display status and operate a physical barrier. The available record does not establish measurement accuracy, suitable threshold selection, decision latency, or repeatability. These are therefore treated as validation questions, not achieved performance.

The primary objective is to document the reported local sensing-and-barrier-control concept and define a reproducible method for determining whether it distinguishes safe from hazardous water levels reliably. The secondary objectives are to: (1) describe the ultrasonic measurement model and its calibration needs; (2) identify implementation details that require confirmation; (3) define metrics for sensor error, false alarms, detection delay, and barrier response; (4) examine environmental, electrical, and operational risks; and (5) specify the additional modules and tests required before the system could be described as network-enabled or field-ready.

System boundary and assumptions. The evaluation target is a bench-scale mock-up, not a live tunnel. The reported hardware comprises an Arduino Uno/ATmega328P, an HC-SR04 sensor, a servo-operated barrier, and a 20 × 4 I²C LCD. Exact part variants, wiring, firmware, thresholds, reference dimensions, sample count, power arrangement, and calibration procedure remain unconfirmed. Cellular/GPS/LoRaWAN communication and pumping are not counted as implemented functions without additional evidence.

IV. PROPOSED SYSTEM ARCHITECTURE

Figure 1 is a functional interpretation of the components named in the project description; it is not a pin-level circuit diagram. The ultrasonic module measures the distance between the transducer and the water surface. The controller converts distance to estimated depth using a calibrated reference dimension and compares the result with configured thresholds. The LCD provides local status feedback, and the servo is described as operating a small barrier in response to the threshold logic. The electrical connections, thresholds, servo travel limits, and reset sequence must be confirmed against the as-built circuit and firmware before reproducibility can be claimed.

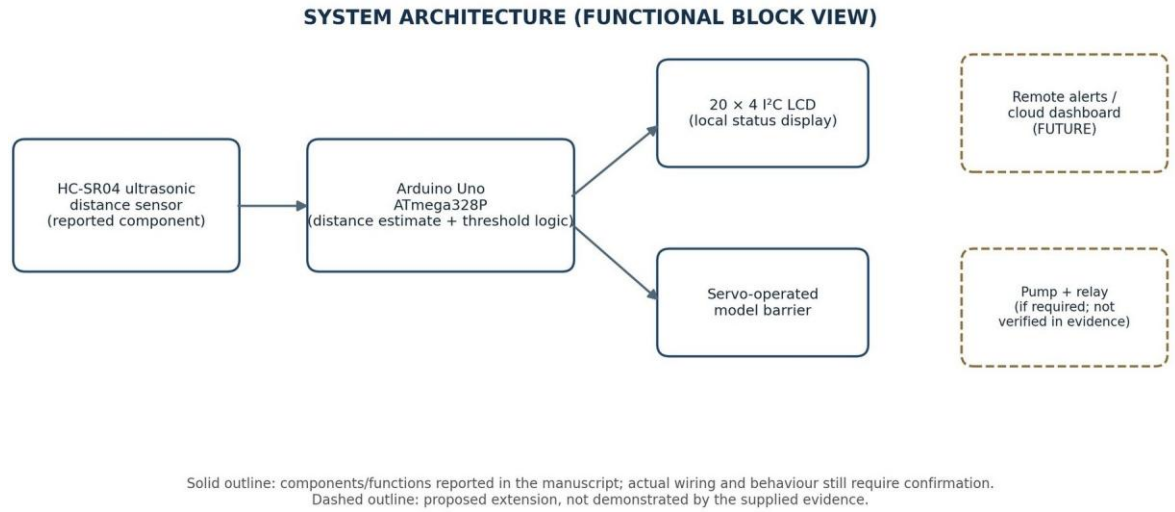


Fig. 1. Functional block view. Solid-outline modules are reported components/functions; dashed-outline modules are proposed extensions not verified by the available evidence.

A production deployment would require additional layers that are not demonstrated here: remote communication, authenticated telemetry, data storage, independent fault detection, protected power and electrical isolation, and possibly a dewatering pump. Networked operation would require a documented communications module and protocol, message schema, retry/timeout rules, and end-to-end delivery tests. A remote control path would also require authorization and a safe local override; connectivity alone is not a reason to enable remote actuation.

V. HARDWARE AND SOFTWARE REQUIREMENTS

Table 2 distinguishes the hardware named in the source description from details that remain to be confirmed. For a standard Arduino Uno R3, the manufacturer specification lists 32 KB flash, 2 KB SRAM, and 1 KB EEPROM [17]. The SRAM figure is important: 32 KB is flash capacity, not SRAM. The exact board revision in the photographed model has not been independently verified.

Table 2. Reported components and evidence status.

Item	Reported function / detail	Evidence status and required confirmation
Arduino Uno / ATmega328P	Local embedded processing. Standard Uno R3: 5 V logic, 16 MHz clock, 32 KB flash, 2 KB SRAM, and 1 KB EEPROM [17].	Named in the source manuscript. Confirm the exact board revision and installed microcontroller.
HC-SR04 ultrasonic module	Non-contact distance measurement using Trigger and Echo timing.	Named in the source. Confirm the actual module, mounting height/orientation, calibrated range, and test geometry.
Servo-operated model barrier	Actuator for the scaled gate/barrier; SG90/MG996R are mentioned as alternatives.	Installed model, supply, travel range, torque, linkage, and end positions are not confirmed.
20 × 4 LCD with I²C interface	Local readout of water-level or system state.	A display is visible in the photographs. Confirm model, I²C address, refresh format, and displayed values.
Pump and high-current relay	Discussed as a possible automatic dewatering subsystem.	Not sufficiently demonstrated by photographs or test records. Do not report operation or efficiency as achieved without evidence.
Hardware override / kill switch	Described as a manual safety override.	Circuit and independent operating test are absent. Confirm wiring and whether it directly isolates relevant power paths.
GSM, GPS, LoRaWAN, cloud interface	Proposed for remote telemetry, alerts, or geotagging.	Future scope only in the supplied record; no identifiable module or data log is provided.
Firmware / development environment	Project description mentions polling, pulse timing, threshold logic, and PWM.	Source code, version, pin map, sample interval, filters, and threshold constants were not supplied.

VI. SYSTEM METHODOLOGY AND WORKING PRINCIPLE

A. Ultrasonic distance and level estimation

In an ultrasonic time-of-flight measurement, the controller triggers the module, which emits an acoustic burst; the Echo signal indicates the round-trip travel time. If t_e is the measured round-trip time and c is the speed of sound, the estimated sensor-to-surface distance is:

$$d = (c \times t_e) / 2 \quad (1)$$

If H is the calibrated vertical distance from the sensor reference point to the floor, water depth h is estimated as:

$$h = H - d \quad (2)$$

These equations assume a stable mounting reference, a sufficiently reflective water surface, correct timing, and a validated measurement range. Temperature affects sound speed; angled surfaces, splashing, foam, condensation, obstructions, multipath echoes, and water motion can disturb readings. The sensor should not be assigned a universal accuracy figure. A ruler or calibrated depth gauge should be used to measure bias and repeatability over the full depth range of the model [5].

B. Decision logic and control sequence

The project description refers to threshold-based control, but numerical thresholds and firmware are unavailable. The following sequence is a reproducible validation baseline, not a reconstruction of the installed firmware: (1) acquire an echo-time sample; (2) reject timeouts and implausible readings; (3) convert a valid distance to estimated depth; (4) display depth and local operating state; (5) compare depth with pre-approved thresholds; (6) command the model barrier to its defined position when a critical threshold is reached; and (7) record the input, decision, command, and observed response. The sequence must be compared with the actual code before being described as implemented.

For a safety-oriented implementation, one reading should not normally be sufficient to trigger an access-control actuator. Plausibility checks and a configurable persistence rule can reduce transient triggers. Hysteresis can limit repeated state changes near a threshold by using a higher level to enter a critical state and a lower reset level to leave it. The reset criterion must follow a site-specific hazard assessment. These are recommendations, not verified features of the present prototype.

C. Recommended validation algorithm

```
INITIALIZE sensor, display, controller, and available actuators
LOAD validated measurement limits and control thresholds
REPEAT at the configured sampling interval:
  echo_time ← acquire_ultrasonic_echo()
  IF echo_time is missing, timed out, or outside the validated range:
    reading_status ← INVALID
    display "SENSOR CHECK"
    record sensor fault and timestamp
    DO NOT interpret invalid data as a safe water level
  ELSE:
    distance ← (speed_of_sound × echo_time) / 2
    depth ← reference_height − distance
    depth ← apply_documented_filter(depth)
    state ← classify_water_level(depth, thresholds, hysteresis)
    update_local_display(depth, state)
    IF state = CRITICAL:
      command barrier to the pre-approved safe position
      record time, raw reading, depth, state, and command
      IF actuator feedback is installed: verify response and log mismatch
END REPEAT
```

Algorithm 1. Proposed sensor-validation and barrier-control baseline; not a reconstruction of the prototype firmware.

The reference height, valid measurement range, sample interval, filter, thresholds, hysteresis settings, and barrier safe position must be defined for the actual hardware. Actuator feedback and fault logging are conditional on being installed. The algorithm must be reconciled with the source code and tested before it can be reported as implemented.

VII. PROTOTYPE IMPLEMENTATION AND PHOTOGRAPHIC EVIDENCE

Figure 2 consolidates four photographs of a white scale model with open channel/tunnel geometry, vertical structures, exposed wires, electronic modules, and LCD-type displays. The photographs document assembly and a team demonstration; they do not establish pin-level wiring, firmware behaviour, calibrated readings, pump performance, environmental protection, remote connectivity, or repeat-trial results.



Fig. 2. Supplied prototype photographs: (a) overall model view; (b) close-up of the structure and exposed wiring; (c) team demonstration; and (d) additional demonstration view. The images document assembly, not quantitative performance.

VIII. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

No raw sensor samples, calibration data, reference measurements, trial counts, measured actuation timings, power records, or communication-delivery logs were supplied. The project description outlines tests for detection latency, barrier actuation, and long-duration operation, but these remain proposed procedures rather than completed experiments. No numerical estimate of detection accuracy, false-alarm rate, response time, water-removal rate, pump efficiency, or uptime can therefore be reported from the available evidence.

A reproducible evaluation should use a measured reference height, a ruler or calibrated depth gauge, controlled changes in water level, and a timestamped log of raw Echo times, estimated depths, state transitions, servo commands, and observed barrier position. Raise and lower the water repeatedly through each threshold. Include slow ramps, rapid changes, surface disturbance, and invalid-echo conditions. Each trial should record firmware version, sensor geometry, water temperature if compensation is used, sample interval, supply voltage, trial count, threshold values, and failures. Initial testing should remain on the scale model; live traffic locations are outside the present scope.

Table 3. Evaluation metrics and evidence required. These are future-test definitions, not measured results.

Metric	Definition / calculation	Required data
Depth error	e_i = estimated depth − reference depth. Report bias and MAE = mean($\|e_i\|$); RMSE may also be reported.	Paired sensor estimates and calibrated reference measurements.
Detection classification	Accuracy = (TP + TN) / (TP + TN + FP + FN). Also report precision and recall for labelled hazardous events.	Independent labelled safe/critical trials; define TP, TN, FP, and FN before testing.
Detection latency	Time from the reference depth crossing a configured threshold to the firmware recording the critical state.	Synchronized reference and controller timestamps.
Barrier actuation latency	Time from the critical-state decision to the barrier reaching its defined safe position.	Decision timestamp and measured physical end-position timestamp.
Repeatability	Distribution of error and latency over repeated trials; include sample count and range or confidence interval.	Repeated trials at stated depths and water-rise rates.
Connectivity / alerts	Delivery ratio = valid messages received / messages sent; record transmission-to-reception delay.	Only applicable after communication hardware is installed and configured.
Pump removal performance	Volumetric removal rate = water volume removed / pumping time, under stated head and supply conditions.	An installed pump, measured volume, test geometry, and pump on/off logs.

The source description mentions candidate targets of less than 5% error and barrier actuation within 3 s. Because the denominator, measurement method, thresholds, and test conditions are not documented, these figures require author confirmation and must not be presented as achieved results.

IX. RESULTS AND DISCUSSION

A. Evidence-supported observations

The available evidence supports three limited observations. First, a physical scale model was assembled and photographed during a demonstration (Fig. 2). Second, the project description identifies the intended local sensing-and-control concept as an HC-SR04/Arduino arrangement connected to an LCD and a servo-operated model barrier. Third, GSM, GPS, LoRaWAN, and centralized monitoring are described as future extensions. These points establish the scope of the design and the existence of a mock-up; they do not establish quantitative operation under controlled conditions.

B. Unverified performance claims

The project description reports up to 95% water-level accuracy and approximately 80% pump efficiency. These claims cannot be independently evaluated because raw readings, reference measurements, trial count, the error definition, pump identification, pump curve, hydraulic head, and measured electrical input/output data were not supplied. They are therefore excluded from the results. The same limitation applies to the described hardware “kill switch” and high-current relay: no circuit diagram or independent operating test is available. Photographs of the mock-up and exposed wiring do not show that a pump or override was installed, powered, or tested safely.

C. Interpretation and research contribution

At the current evidence level, the main outcome is a documented prototype concept accompanied by a test plan that makes the next research step explicit. The system has plausible educational value as a tangible demonstration of sensor-to-controller-to-actuator integration. A stronger scientific contribution requires calibration and repeated trials: compare estimated depth with a reference across the operating range, measure barrier decisions against threshold-crossing events,

and deliberately test failure cases. Even a single successful demonstration would not establish reliability during a real flood.

A direct performance ranking against related work is inappropriate because the cited studies use different sensors, geometries, communication layers, and validation procedures. Future comparison should use common metrics such as mean absolute or root-mean-square depth error, event-level recall for hazardous states, median and worst-case actuation delay, and message-delivery ratio. Every result should include sample size and test conditions.

X. SECURITY, SAFETY, AND LIMITATIONS

A. Sensor failure and unreliable readings

An ultrasonic sensor can return invalid or unstable distances because of range limits, geometry, splashing, foam, condensation, or obstruction. Firmware should distinguish invalid data from a safe water level, expose a sensor-fault state, and record such events. Periodic plausibility checks and redundant sensing may be warranted in a deployed installation, but are not documented as current features.

B. Actuator and access-control safety

A servo-operated model barrier is not equivalent to a traffic-rated barricade. A real barrier requires assessment of force/torque, entrapment, obstruction, mechanical loads, and failure position. A gate that closes after a sensor or controller fault could itself introduce risk. Independent safety logic, manual supervision, a defined safe state, emergency egress, and jurisdiction-specific review are required before live use. The claimed “kill switch” must be tested and documented; a switch connected only to a microcontroller reset pin would not necessarily provide the independent pump/gate isolation implied by the description.

C. Water, electrical protection, and cybersecurity

Exposed electronics and wiring visible in the prototype photographs are acceptable only for a controlled bench-scale demonstration. A real installation requires appropriately rated enclosures, cable glands, protected supplies, electrical separation from pumps and actuators, suitable residual-current and overcurrent protection, grounding/bonding where required, corrosion-resistant connectors, drainage, and maintenance access. No enclosure rating, electrical certification, redundancy, or backup power is documented for this model.

No remote link is verified in the supplied prototype record, so remote alerts, geotagging, and centralized monitoring remain future-work items. If connectivity is added, device identity, authentication, encrypted communication where supported, access-control policy, controlled firmware updates, secure configuration, logging, and offline-safe behaviour should be designed from the outset [18]. Loss of cellular or internet service must not be treated as evidence that a site is safe.

D. Scope limitations

This paper is limited by the absence of source code, unconfirmed part variants and circuit details, missing calibration and raw experimental data, lack of quantitative testing, uncertain pump/relay/override implementation, no confirmed communications module, and no full-scale field deployment. Model results cannot by themselves establish performance in an underground tunnel, where airflow, mounting height, surface reflections, vibration, electromagnetic interference, corrosion, debris, and water-inflow dynamics differ materially.

XI. CONCLUSION AND FUTURE SCOPE

This paper documents an Arduino-based prototype concept for waterlogging detection and barrier control in a scaled subterranean-infrastructure model. The project description identifies an HC-SR04 ultrasonic sensor, an Arduino Uno/ATmega328P, a local I²C LCD, and a servo barrier; the photographs support the existence of a physical mock-up and its demonstration. The available evidence does not verify internet connectivity, measured sensor accuracy, automatic pump dewatering, the stated pump efficiency, an independent hardware override, or performance under repeated controlled tests. These gaps materially limit the conclusions that can be drawn.

The next development stage should provide source code or an archived firmware listing, an as-built circuit diagram and bill of materials, exact component identifiers, mounting dimensions, calibrated sensor readings, repeated threshold-crossing tests, measured actuation timing, and failure-mode tests. Pump integration and remote telemetry should be evaluated only after these fundamentals are documented. A future connected version could add GSM, Wi-Fi, or LoRaWAN and a dashboard, while retaining locally safe behaviour during communication failures. Multi-sensor checks, water-rise-rate estimation, backup power, environmental enclosures, maintenance diagnostics, and supervised field trials are reasonable extensions. Operational-readiness claims should be reserved for a system tested under representative conditions and reviewed against applicable safety requirements.

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Conflict of interest — AUTHOR TO CONFIRM.

Author contributions — AUTHOR TO CONFIRM the contribution of each named author.

Data availability — No raw experimental dataset, firmware source, or calibration record was supplied for this reconstruction. AUTHOR TO PROVIDE a repository/DOI or final data-availability statement if data exist.

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Editorial note: *Before submission, confirm the declarations above and verify all bibliographic details against the original publications and the target venue's formatting rules. No new experimental results have been added.*